

time. The following spreadsheet shows the first and final three monthly cash flows.

Month	Total Monthly Payment	Monthly Interest Payment	Monthly Principal Repayment	Remaining Principal
1	\$ 4,417.63	\$ 3,500.00	\$ 917.63	\$ 799,082.37
2	\$ 4,417.63	\$ 3,495.99	\$ 921.64	\$ 798,160.73
3	\$ 4,417.63	\$ 3,491.95	\$ 925.68	\$ 797,235.05
...				
358	\$ 4,417.63	\$ 57.48	\$ 4,360.15	\$ 8,777.61
359	\$ 4,417.63	\$ 38.40	\$ 4,379.23	\$ 4,398.39
360	\$ 4,417.63	\$ 19.24	\$ 4,398.39	\$ 0.00

Equity Instruments and the Time Value of Money

Equity investments, such as preferred or common stock, represent ownership shares in a company which entitle investors to receive any discretionary cash flows in the form of dividends. Unlike fixed-income instruments, equity investments have no maturity date and are assumed to remain outstanding indefinitely, or until a company is sold, restructured, or liquidated. One way to value a company's shares is by discounting expected future cash flows using an expected rate of return (r). These cash flows include any periodic dividends received plus the expected price received at the end of an investment horizon.

Common assumptions associated with valuing equity instruments based upon dividend cash flows often follow one of three general approaches:

- **Constant Dividends:** An investor pays an initial price (PV) for a preferred or common share of stock and receives a fixed periodic dividend (D).
- **Constant Dividend Growth Rate:** An investor pays an initial price (PV) for a share of stock and receives an initial dividend in one period (D_{t+1}), which is expected to grow over time at a constant rate of g .
- **Changing Dividend Growth Rate:** An investor pays an initial price (PV) for a share of stock and receives an initial dividend in one period (D_{t+1}). The dividend is expected to grow at a rate that changes over time as a company moves from an initial period of high growth to slower growth as it reaches maturity.

The simplest case of a stock that is assumed to pay constant dividends in perpetuity is shown in Exhibit 4.